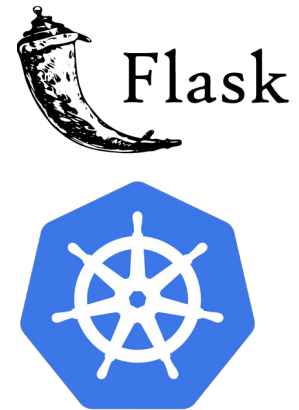
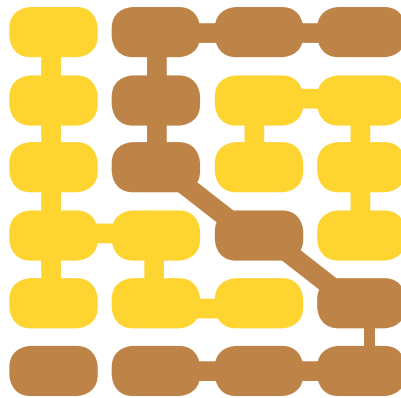


Python Microservices Using Flask on Kubernetes

Microservices follow domain driven design (DDD), irrespective of the platform on which they are developed. Python microservices are not an exception. The object-oriented features of Python 3 make it even easier to model services on the lines of DDD. Part 10 of this series demonstrates how to deploy *FindService* of user management system as a Python microservice on Kubernetes.



The power of microservices architecture lies partly in its polyglot nature. The enterprises decompose their functionality into a set of microservices and each team chooses a platform of its choice.

Our user management system was already decomposed into four microservices, namely, *AddService*, *FindService*, *SearchService* and *JournalService*. The *AddService* was developed on the Java platform and deployed on the Kubernetes cluster for resilience and scalability. This doesn't mean that the remaining services are also developed on Java. We are free to choose a suitable platform for individual services.

Let us pick Python as the platform for developing the *FindService*. The model for *FindService* has already been designed (refer to the March 2022 issue of *Open Source For You*). We only need to convert this model into code and configuration.

The Pythonic approach

Python is a general-purpose programming language that has been around for about three decades. In the early days, it

was the first choice for automation scripting. However, with frameworks like Django and Flask, its popularity has been growing and it is now adopted in various other domains like enterprise application development. Data science and machine learning pushed the envelope further and Python is now one of the top three programming languages.

Many people attribute the success of Python to its ease of coding. This is only partially true. As long as your objective is to develop small scripts, Python is just like a toy. You love it. However, the moment you enter the domain of serious large-scale application development, you will have to handle lots of ifs and buts, and Python becomes as good as or as bad as any other platform. For example, take an object-oriented approach! Many Python developers may not even be aware of the fact that Python supports classes, inheritance, etc. Python does support full-blown object-oriented development, but in its own way — the Pythonic way! Let us explore it!

The domain model

AddService adds the users to the system by saving the data into a MySQL database. The objective of the *FindService*